

جامعة الأميرة
نورة بنت عبدالرحمن
كلية علوم الحاسب والمعلومات



**CS313 Advanced Programming Language
Computer Science Department**

Term 1, 2025-2026

Project B: Hotel Reservation System

Project Objectives

After implementing the term project, students will be able to:

- Apply the core principles of design, implementation of object-oriented software solutions
- Implement object-oriented software solutions that combine JavaFx API with database programming
- Collaborate effectively in teams and write technical reports.

Project Description

The objective of this project is to design and implement a Hotel Reservation System using JavaFX API. It provides role-based access for guests, receptionists, and administrators through a login interface. Guests can browse available rooms, make reservations, cancel bookings, and view past stays. Receptionists can confirm reservations, manage check-ins and check-outs, and update room availability. Administrators oversee all reservations, manage user accounts, configure hotel services, and generate revenue reports. The system uses a MySQL database to store guest information, room details, and booking history. JavaFX should be used to create a user-friendly interface, with controllers connecting the front-end to the back-end database.

Project Requirements

In this project, you must use the concepts of CS 313: Inheritance, Interface, Composition, exception handling, GUI, Database, collections and generic collections

Project Submission

- All teams have to submit their projects on week 12 by the end of your class time.
- You have to submit:
 - The source code for your project.
 - A Report that includes:
 - Introduction: a brief overview of your project.
 - System hierarchy: represents user navigation hierarchy.
 - UML class diagram: represents all classes and their relationships
 - Database schema
 - Snapshot of the execution trace.

- Snapshot of the code.
- Teams' discussion will be on week 13